

ML5.js: Machine Learning Made More User-Friendly



Machine learning is everywhere these days. No wonder people from diverse backgrounds want to acquire ML skills. However, the interfacing is complex for those without proper knowhow of certain underlying concepts. There are many libraries and frameworks that have recently made ML more user-friendly. ML5.js is one such library that enables you to get started with machine learning without many prerequisites. Let's explore this wonderful ML library called ML5.js.

The traditional linear and micromanaged approach of the software development process has been challenged by the emergence of machine learning (ML). The ML based approach has fundamentally changed the manner in which software is developed. The traditional approach required the software developer to provide a micro-level sequence of instructions, which was a big barrier in solving certain types of problems such as object recognition, speech recognition, etc. By using the ML approach, solving these problems has become a possibility, with a manageable level of complexity.

As machine learning is getting more and more popular across a spectrum of domains, users from diverse backgrounds want to acquire ML skills.

As some of them may like formal training in subjects such as probability and statistics, there is a need for libraries that can make the process look simpler by hiding the underlying complexities. Recently, a few such powerful libraries have evolved – TensorFlow and Keras, to name a few. To this list, we can now add ML5.js as well.

ML5.js

The primary objective of ML5.js is to make ML user-friendly and accessible to a diverse audience. ML5.js adds another layer on top of TensorFlow.js to make it even simpler. The primary advantage of ML5.js is that it can be used in the browser itself without the requirement to install complicated dependencies. The funding and support to ML5.js has been given by the Google Education Grant at NYU's ITP/IMA programme.

ML5.js is available as open source software. It provides a high level interface to the powerful TensorFlow.js library. If you have some exposure to TensorFlow.js, you might be aware that it is both easy to use and very powerful. As ML5.js has added another layer on TensorFlow.js, it should be even simpler.

The browser based approach is one of the important attributes of ML5.js. An excellent video that introduces ML5.js is available at <https://youtu.be/jmznx0Q1fP0>. It will certainly help you to get started with ML5.js, as the official documentation itself has recommended this video.

ML5.js requires no installation as it is browser based. To get started, you can right away open a code editor, start typing the code and execute it in a Web browser to see ML in action. Let's explore each of the steps to get started with ML5.js.